The Structure of c-Concave Functions
Envelopes, ordinary convexity, cones, and cross-curvature

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Abstract

For an arbitrary cost, c-concave functions form a lower-envelope class: they are closed under pointwise infima and addition of constants, hence form a min-plus, or tropical, cone. They need not form an ordinary convex set. This note identifies the additional structures that produce ordinary convexity. For the bilinear cost $c(x, y) = -\langle x, y\rangle$, one recovers the cone of closed concave functions. For $c(x, y) = |x - y|^2/(2\tau)$, one obtains the convex, but non-conic, set of τ-semiconcave functions. For the metric cost $c = Ld$, one obtains the closed unit ball $\mathrm{Lip}(u) \leq L$, again convex but not a cone. For smooth twisted costs, the theorem of Figalli–Kim–McCann characterizes convexity of the c-concave class by nonnegative cross-curvature, a strengthening of the weak Ma–Trudinger–Wang condition. This explains the nontrivial convexity result for squared geodesic distance on round spheres and products of round spheres. We also summarize Bregman, quotient, Wasserstein-lifted, matrix, information-geometric and Gromov–Wasserstein examples, and distinguish ordinary convexity from the universal tropical structure and from quantitative strong c-concavity.

1 Conventions and the universal structure

The answer depends critically on the sign convention, the domains, and whether "convex" refers to ordinary linear combinations or to min-plus combinations. We use throughout the convention of the OT4ML book.

1.1 Transform, fixed points, and supports

Let X and Y be sets and let $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$. For clarity, most statements below are written for finite-valued functions and costs; the same identities hold for extended-real-valued functions whenever undefined combinations of infinities are excluded.

Definition 1.1 (c-transform and c-concavity)**.** For $f : X \to \mathbb{R} \cup \{-\infty\}$ and $u : Y \to \mathbb{R} \cup \{-\infty\}$, define

$$f^c(y) := \inf_{x\in X}\{c(x, y) - f(x)\}, \qquad u^{\bar c}(x) := \inf_{y\in Y}\{c(x, y) - u(y)\}.$$

A function u on Y is c-concave if $u = f^c$ for some f on X. We denote the resulting class by $\mathfrak{C}_c(Y)$. The c-subdifferential is

$$\partial^c u(y) := \{x \in X : u(y) + u^{\bar c}(x) = c(x, y)\}.$$

Its elements index the cost sections that support the lower envelope at y.

Thus every c-concave function is an infimum of shifted cost sections,

$$u(y) = \inf_{x\in X}\{c(x, y) - f(x)\}.$$

These sections play the role of affine supports in ordinary concave analysis.

Proposition 1.2 (Closure by double transformation)**.** *The transforms reverse order. Moreover,*

$$f \le f^{c\bar{c}}, \qquad f^{c\bar{c}c} = f^c.$$

Consequently, u is c-concave if and only if $u = u^{\bar{c}c}$. In that case $u^{\bar{c}c}$ is the smallest c-concave majorant of an arbitrary function u.

Proof. If $f_0 \le f_1$, then $c - f_0 \ge c - f_1$, hence $f_0^c \ge f_1^c$. Also, $f^c(y) \le c(x,y) - f(x)$ for every (x,y), so $f(x) \le c(x,y) - f^c(y)$ and therefore $f \le f^{c\bar{c}}$. Applying the order-reversing transform gives $f^c \ge f^{c\bar{c}c}$, while the same majorization applied to f^c gives the reverse inequality. This proves the triple-transform identity and the fixed-point characterization.

Finally, if h is c-concave and $h \ge u$, then $h^{\bar{c}} \le u^{\bar{c}}$. Transforming again and using $h = h^{\bar{c}c}$ gives $h \ge u^{\bar{c}c}$. □

1.2 The min-plus cone that is always present

Ordinary convexity is exceptional, but a nonlinear analogue of conic structure is automatic.

Proposition 1.3 (Universal min-plus structure)**.** *Let $(u_i)_{i\in I} \subset \mathfrak{C}_c(Y)$ and $(a_i)_{i\in I} \subset \mathbb{R}$. Then, whenever the right-hand side is proper,*

$$y \longmapsto \inf_{i\in I}\{u_i(y) + a_i\} \quad \textit{belongs to } \mathfrak{C}_c(Y).$$

In particular, $\mathfrak{C}_c(Y)$ is closed under arbitrary pointwise infima and under addition of constants.

Proof. Write $u_i = f_i^c$. Since $(f_i - a_i)^c = u_i + a_i$, setting $F(x) = \sup_i\{f_i(x) - a_i\}$ gives directly

$$\inf_i(u_i + a_i) = \inf_{i,x}\{c(x,\cdot) - f_i(x) + a_i\} = \inf_x\{c(x,\cdot) - F(x)\} = F^c.$$

Thus the lower envelope is a c-transform for the original cost on the original domain X. □

Remark 1.4 (Tropical, not ordinary, convexity). In min-plus algebra, "addition" is pointwise minimum and scalar "multiplication" is addition of a constant. Proposition 1.3 says that every c-concave class is a tropical convex cone. This statement does not imply closure under the ordinary average $(u_0 + u_1)/2$.

Example 1.5 (A finite cost for which ordinary convexity fails)**.** Take $X = \{1,2\}$, $Y = \{1,2,3\}$ and

$$C = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}.$$

The two rows $u_0 = (0,0,2)$ and $u_1 = (0,2,0)$ are c-concave: each is obtained by suppressing the other row with a sufficiently negative source potential. Their average $\bar{u} = (0,1,1)$ is not c-concave. Indeed, every finite c-concave vector has the form

$$\big(\min\{A,B\}, \min\{A, B+2\}, \min\{A+2, B\}\big).$$

If its first coordinate is zero, then $A, B \ge 0$ and at least one of them is zero. If $A = 0$, its second coordinate is zero; if $B = 0$, its third coordinate is zero. Neither possibility gives $(0,1,1)$.

The following elementary obstruction explains why ordinary cone structure is rare for costs on compact spaces.

Proposition 1.6 (A uniform modulus obstructs conic scaling)**.** *Assume all sections $y \mapsto c(x,y)$ are L-Lipschitz, with the same finite L. Then every finite $u \in \mathfrak{C}_c(Y)$ is L-Lipschitz. If the class contains a nonconstant function, it cannot be an ordinary convex cone.*

Proof. An infimum of functions sharing the same Lipschitz constant is L-Lipschitz. If u is nonconstant and $\lambda > L/\mathrm{Lip}(u)$, then λu has Lipschitz constant larger than L and therefore cannot be c-concave. □

2 Three model costs

The bilinear, quadratic, and metric costs give three distinct answers: a cone, an affine convex class, and a norm ball.

2.1 Bilinear cost: the cone of concave functions

Theorem 2.1 (Bilinear cost). *Let $X = Y = \mathbb{R}^d$ and $c(x, y) = -\langle x, y\rangle$. Then the proper upper-semicontinuous c-concave functions are exactly the proper upper-semicontinuous concave functions. The finite-valued part of this class is an ordinary convex cone, with the constants as a lineality subspace.*

Proof. If $u = f^c$, then

$$-u(y) = \sup_x\{\langle x, y\rangle + f(x)\} = (-f)^*(y),$$

so $-u$ is closed convex. Conversely, if $h = -u$ is proper lower-semicontinuous convex, Fenchel–Moreau gives $h = h^{**}$ (Rockafellar, 1970). Choosing $f = -h^*$ yields

$$f^c(y) = \inf_x\{-\langle x, y\rangle + h^*(x)\} = -h^{**}(y) = u(y).$$

Nonnegative linear combinations of finite concave functions are concave, which gives the cone statement. □

Remark 2.2 (Restricted slope domains). The full-space hypothesis matters. If X is restricted, the admissible supporting slopes of u lie in $-X$. When X is convex, the resulting class is still convex under standard support/closedness assumptions, but it is a cone only if the allowed slope set is itself stable under nonnegative scaling. Thus even the bilinear cost need not produce a cone on bounded domains.

2.2 Quadratic cost: semiconcavity at a fixed scale

Let $c_\tau(x, y) = |x - y|^2/(2\tau)$ with $\tau > 0$, and set $q_\tau(y) = |y|^2/(2\tau)$.

Theorem 2.3 (Quadratic cost). *On $X = Y = \mathbb{R}^d$, a proper upper-semicontinuous function u is c_τ-concave if and only if $u - q_\tau$ is concave. Equivalently, in the distributional sense,*

$$D^2u \preccurlyeq \tau^{-1}I.$$

The class is an ordinary convex set, but not an ordinary cone.

Proof. Expanding the square gives

$$f^{c_\tau}(y) - q_\tau(y) = \inf_x\left\{\frac{|x|^2}{2\tau} - f(x) - \frac{\langle x, y\rangle}{\tau}\right\},$$

an infimum of affine functions of y, hence a closed concave function. The converse follows from Theorem 2.1 after rescaling the slope. The set is convex because $u \mapsto u - q_\tau$ is affine. It is not a cone: $u = q_\tau$ belongs to the class, whereas $2u - q_\tau = q_\tau$ is not concave. □

This is the first warning against using "convex cone" as a generic description. The curvature scale τ^{-1} is fixed by the cost. Addition or large positive scaling changes that scale.

Remark 2.4 (Euclidean balls and products of balls). On a convex Euclidean ball, intrinsic geodesic distance is ordinary Euclidean distance. The squared-distance cost is therefore the restriction of the quadratic model above. The c-concave class remains convex, with the additional constraint that supporting slopes come from the prescribed source ball; it is not a cone. A product of Euclidean balls gives the same statement in the product Euclidean space. This elementary case should not be confused with the nontrivial theorem for products of round spheres in Section 4.

2.3 Metric cost: a Lipschitz ball, not a cone

Theorem 2.5 (Metric cost)**.** *Let (X, d) be any metric space, let $L > 0$, and set $c(x, y) = Ld(x, y)$. Then*

$$\mathfrak{C}_c(X) = \{u : X \to \mathbb{R} : \mathrm{Lip}(u) \le L\}.$$

For every such u, one has $u^c = -u$. This class is closed, convex, balanced, and stable under pointwise minimum and maximum, but it is not a cone unless it contains only constants.

Proof. Every section $y \mapsto Ld(x, y)$ is L-Lipschitz, so every c-transform is L-Lipschitz. Conversely, if $\mathrm{Lip}(u) \le L$, then $u(x) \le u(y) + Ld(x, y)$, hence $Ld(x, y) - u(x) \ge -u(y)$. Taking $x = y$ shows that the infimum is exactly $u^c(y) = -u(y)$. Applying the same identity to $-u$ gives $u = (-u)^c$. The structural assertions follow from elementary properties of the Lipschitz seminorm; non-conicity also follows from Proposition 1.6. □

Modulo additive constants, this is the closed radius-L ball of the normed space defined by the Lipschitz seminorm. The union over all $L < \infty$ is the vector space of Lipschitz functions, but changing L changes the cost. In particular, for the plain geodesic cost $c = d_M$ on a sphere or on any other metric space, no curvature theorem is needed: the answer is always the 1-Lipschitz ball.

2.4 What uniform semiconcavity does and does not imply

The quadratic calculation has a useful one-sided generalization.

Proposition 2.6 (Semiconcavity inherited from cost sections)**.** *Let $Y \subset \mathbb{R}^d$ be convex. Suppose that for every $x \in X$, the function*

$$y \longmapsto c(x, y) - \frac{\Lambda}{2}|y|^2$$

is concave. Then every finite c-concave function is Λ-semiconcave. In particular, for $c(x, y) = h(x - y)$ it suffices that $D^2 h \preccurlyeq \Lambda I$.

Proof. Subtracting $\Lambda|y|^2/2$ from a c-transform leaves an infimum of concave functions. Pointwise infima preserve concavity because their hypographs are intersections of convex sets. □

This proposition gives regularity of individual potentials. It does not, by itself, show that the whole c-concave class is closed under ordinary convex combinations. Quadratic costs are special because the shifted class is exactly the full concave cone.

3 Ordinary convexity and nonnegative cross-curvature

The preceding examples suggest that ordinary convexity is a property of the cost, not a formal consequence of the transform. In smooth finite dimensions, the exact criterion is nonnegative cross-curvature.

3.1 The differential condition

Assume X and Y are smooth d-dimensional domains, $c \in C^4(X \times Y)$, the mixed Hessian $D^2_{xy}c$ is invertible, and the usual twist and domain c-convexity hypotheses hold. A c-segment $s \mapsto x_s$ with base $\bar{y}$ is defined by requiring

$$D_y c(x_s, \bar{y}) = (1 - s)D_y c(x_0, \bar{y}) + sD_y c(x_1, \bar{y}).$$

For compatible c-segments x_s and y_t, define the cross-curvature by

$$\mathrm{cross}_c(\dot{x}_0, \dot{y}_0) := -\left.\frac{\partial^4}{\partial s^2 \partial t^2} c(x_s, y_t)\right|_{s=t=0}. \qquad (1)$$

The cost has *nonnegative cross-curvature* when this quantity is nonnegative for all directions. The weak MTW condition A3w imposes the same sign only on null directions satisfying the corresponding mixed-orthogonality condition. Nonnegative cross-curvature is therefore stronger than A3w. For background on these hypotheses and their role in optimal-transport regularity, see Villani (2009) and Kim and McCann (2010).

Theorem 3.1 (Figalli–Kim–McCann)**.** *Under the smooth twist, nondegeneracy, compactness, and domain c-convexity hypotheses above, the set $\mathfrak{C}_c(Y)$ of c-concave functions is an ordinary convex set if and only if c has nonnegative cross-curvature (Figalli et al., 2011).*

Mechanism of the theorem. At a point y_0, choose supporting cost sections for two potentials u_0 and u_1, indexed by x_0 and x_1. The c-segment x_s based at y_0 interpolates their cost slopes. Nonnegative cross-curvature says that the intermediate sliding mountain lies below the same convex combination of the endpoint mountains. It therefore supports $u_s = (1-s)u_0 + su_1$ at y_0. Repeating this at every point proves c-concavity of u_s. Conversely, applying convexity of the potential class to pairs of individual mountains recovers the cross-curvature inequality.

Remark 3.2 (A3w is not enough). A3w, or Loeper's maximum principle, controls the maximum of two endpoint mountains and is central to continuity of optimal maps. Convexity of the entire potential class requires the stronger "below the chord" inequality, hence nonnegative cross-curvature in all directions. Regularity of optimal maps and ordinary convexity of all potentials are related but distinct properties.

Remark 3.3 (Local supports versus global c-concavity). There is a second structural role for MTW-type assumptions. Under twist, appropriate c-convexity of the domains, and A3w, a potential admitting the correct local c-supports can be promoted to a globally c-concave potential. Loeper and Trudinger (2021) give a geometric proof under only twice-differentiable costs. This local-to-global principle concerns recognition of one potential; Theorem 3.1 concerns closure of the whole class under ordinary averages and requires the stronger NNCC condition.

3.2 A nonsmooth formulation

The smooth tensor is not needed to state the underlying geometry. A path, which need not be continuous, x_s from x_0 to x_1, based at $\bar{y}$, is a variational c-segment if

$$c(x_s, \bar{y}) - c(x_s, y) \le (1-s)\big[c(x_0, \bar{y}) - c(x_0, y)\big] + s\big[c(x_1, \bar{y}) - c(x_1, y)\big] \tag{2}$$

for every y. A cost space is NNCC if such a path exists for every admissible triple $(x_0, x_1, \bar{y})$. This definition applies to nonsmooth and infinite-dimensional spaces.

For arbitrary extended costs, the precise convex class is

$$\mathfrak{C}_c^{\partial}(Y) := \{u : \partial^c u(y) \neq \varnothing \text{ at every point where } u(y) \text{ is finite}\}.$$

Theorem 3.4 (Synthetic characterization)**.** *A cost space is NNCC if and only if $\mathfrak{C}_c^{\partial}(Y)$ is closed under ordinary convex combinations. For finite continuous costs on compact spaces, every finite c-concave function has a supporting section, and this reduces to ordinary convexity of $\mathfrak{C}_c(Y)$ (Léger et al., 2025).*

The qualification by nonempty c-subdifferentials matters on noncompact spaces and for costs taking the value $+\infty$. It should not be silently omitted when using the modern infinite-dimensional theorem.

4 Squared geodesic costs on spheres and products

The nontrivial geometric example alluded to in the question concerns squared geodesic distance, not geodesic distance itself.

Theorem 4.1 (Round spheres and their products)**.** *Let*

$$M = \prod_{k=1}^{N} \mathbb{S}_{r_k}^{d_k}$$

be a finite product of round spheres with its product Riemannian metric, and let $c(x, y) = d_M(x, y)^2/2$. Then the finite c-concave functions on M form an ordinary convex set. Unless M is a singleton, this set is not an ordinary cone.

Proof and attribution. The round sphere has nonnegative cross-curvature for squared geodesic distance (Kim and McCann, 2012). Direct products preserve nonnegative cross-curvature, so the same holds for M. The cut locus prevents a naive global use of the smooth formula (1). For one sphere, Sei (2013) give a direct global proof of convexity. For products, the product stability of NNCC combined with the synthetic characterization in Theorem 3.4 gives the stated conclusion. The detailed cut-locus and regularity analysis for multiple products of spheres is developed by Figalli et al. (2013).

Since M is compact, the sections $y \mapsto d_M(x, y)^2/2$ have a common Lipschitz bound $\operatorname{diam}(M)$. Proposition 1.6 then rules out conic scaling as soon as a nonconstant potential is present. □

Remark 4.2 (What is due to Figalli and collaborators). The general equivalence between convexity of the potential class and nonnegative cross-curvature is due to Figalli, Kim and McCann. Convexity on a single sphere was also proved directly by Sei. Kim and McCann established the cross-curvature and product/submersion mechanisms, while Figalli, Kim and McCann treated global regularity on arbitrary finite products of round spheres. Thus "products of spheres" is the likely intended reference. Products of Euclidean balls are already covered by the quadratic Euclidean calculation.

5 Other useful convex classes

Nonnegative cross-curvature is useful partly because it is stable under several constructions. This produces examples far beyond Euclidean or spherical costs.

5.1 Bregman divergences

Proposition 5.1 (Bregman costs are NNCC)**.** *Let E be a Banach space, let $F : E \to \mathbb{R}$ be convex and Gateaux differentiable, and set*

$$D_F(x, y) = F(x) - F(y) - DF(y)[x - y].$$

If $X \subset E$ is convex and $Y \subset E$ is arbitrary, then $(X \times Y, D_F)$ is NNCC. Consequently, the supported D_F-concave class is convex. For the reversed divergence $D_F(y, x)$, the same conclusion holds if $DF(X)$ is convex in E^*.

Proof. For $x_s = (1 - s)x_0 + sx_1$, direct expansion gives

$$D_F(x_s, \bar{y}) - D_F(x_s, y) = (DF(y) - DF(\bar{y}))[x_s] + F(y) - F(\bar{y}) + DF(\bar{y})[\bar{y}] - DF(y)[y],$$

which is affine in s. Hence (2) holds with equality. For the reversed divergence, interpolate linearly in the dual coordinate $DF(x_s)$. □

This includes squared Hilbert distance and explains why relative entropy, viewed as an infinite-dimensional Bregman divergence, has the same structural property.

5.2 Products, quotients, and pullbacks

Proposition 5.2 (Closure mechanisms for NNCC costs)**.** *The following operations preserve nonnegative cross-curvature.*

(i) If each $(X_k \times Y_k, c_k)$ is NNCC, then the product cost $c(x, y) = \sum_k c_k(x_k, y_k)$ is NNCC.

(ii) If $c(x, y) = \|F(x) - G(y)\|_H^2$ for a Hilbert space H and $F(X)$ is convex, then c is NNCC.

(iii) Suitable cost submersions, including smooth Riemannian submersions for squared-distance costs, preserve NNCC.

The first statement explains products of spheres. The third generates quotient geometries, including complex projective spaces from Hopf fibrations and the Bures–Wasserstein geometry of positive semidefinite matrices from the quotient $M \mapsto MM^\top$ of a Hilbert space of matrices (Kim and McCann, 2012; Léger et al., 2025).

5.3 Lifting the cost to probability measures

For a lower-semicontinuous cost c define its Kantorovich lift

$$\mathcal{T}_c(\alpha, \beta) := \inf_{\pi \in \Pi(\alpha,\beta)} \int c(x, y)\, \mathrm{d}\pi(x, y).$$

Theorem 5.3 (Wasserstein lifting of NNCC)**.** *Under the standard Polish-space, lower-semicontinuity, boundedness, and finiteness hypotheses, if $(X \times Y, c)$ is NNCC, then*

$$\big(\mathcal{P}(X) \times \mathcal{P}(Y), \mathcal{T}_c\big)$$

is NNCC. Under a mild condition ensuring finite competitors, the converse also holds (Léger et al., 2025).

For $c(x, y) = |x - y|^2$, this says that the cost $\mathcal{T}_c = W_2^2$ on the Wasserstein space is itself NNCC. Thus an appropriate supported class of W_2^2-concave *functionals of measures* is convex. This is a genuinely infinite-dimensional extension of the quadratic Euclidean statement, and not merely a reformulation of displacement convexity.

5.4 A modern catalogue

The synthetic theory of Léger et al. (2025) establishes NNCC for the following costs, among others. The convexity conclusion in the last column always refers to the supported class $\mathfrak{C}_c^\partial$; it applies to the full finite c-concave class when compactness and finiteness make supports automatic.

Space or cost	Representative formula	Structural source
Hilbert/Bures–Wasserstein	$\mathrm{BW}^2(A, B) = \operatorname{tr} A + \operatorname{tr} B - 2\operatorname{tr}\big((A^{1/2}BA^{1/2})^{1/2}\big)$	Hilbert quotient / cost submersion
Relative entropy	$\mathrm{KL}(\alpha \mid \beta) = \int \log(\mathrm{d}\alpha/\mathrm{d}\beta)\,\mathrm{d}\alpha$	Bregman and mixture geometry
Hellinger and Fisher–Rao	$H^2(\alpha, \beta)$ and $\mathrm{FR}^2(\alpha, \beta)$	Spherical/Hilbert lifting
Selected Hellinger–Kantorovich costs	Unbalanced transport over an NNCC cone	Cost submersion; additional hypotheses are needed
2-Gromov–Wasserstein	GW_2^2 on gauged metric-measure spaces	Lifted quadratic structure
Wasserstein-on-Wasserstein	W_2^2 on $\mathcal{P}_2(X)$ when the base is NNCC	Kantorovich lifting

These examples show that convexity of c-concave potentials is not restricted to costs on vector spaces. They also reinforce the need to distinguish a convex set from a cone: each squared metric carries a fixed curvature or regularity scale, so positive scaling is generally unavailable.

6 Discrete and quantitative refinements

Two further viewpoints are useful in computation and stability theory.

6.1 Finite costs as tropical polytopes

For a matrix $C \in \mathbb{R}^{n\times m}$, a c-concave vector is

$$u_j = \min_{1\le i\le n}\{C_{ij} - f_i\}.$$

Hence the discrete class is the min-plus span of the rows of C. It is a tropical polytope after quotienting by additive constants. The active indices

$$\partial^C u(j) = \{i : u_j = C_{ij} - f_i\}$$

are the discrete c-subdifferential. Example 1.5 shows that a tropical polytope need not be an ordinary convex polytope. This viewpoint is useful for discrete dual potentials, Laguerre cells, and active-set algorithms.

6.2 Strong c-concavity

Ordinary c-concavity only asks that every point admit a supporting cost section. A quantitative version asks that contact be isolated with a prescribed growth modulus.

Definition 6.1 (Strong c-concavity)**.** A c-concave function u is strongly c-concave with modulus $\omega : [0,\infty) \to [0,\infty)$ if, for every y, every $x \in \partial^c u(y)$, and every z,

$$c(x,z) - u(z) - \big(c(x,y) - u(y)\big) \ge \omega(d_Y(y,z)).$$

When $\omega(r) > 0$ for $r > 0$, a supporting section has a unique contact point. For the quadratic cost, this is the strong-concavity inequality for the shifted potential $u - q_\tau$. Under twist and MTW-type hypotheses, differential criteria for strong c-concavity yield quantitative stability of optimal maps and plans (Gallouët et al., 2026). This is a different strengthening from ordinary convexity of the *set* of all potentials: it quantifies the geometry of one potential.

7 Summary and diagnostic checklist

The main answers can be compressed as follows.

Cost / setting	c-concave class	Convex?	Cone?
Arbitrary c	Lower envelopes of shifted sections	Tropical only	Tropical
$-\langle x,y\rangle$ on full spaces	Closed concave functions	Yes	Yes
$\lvert x-y\rvert^2/(2\tau)$	$u - \lvert\cdot\rvert^2/(2\tau)$ concave	Yes	No
$Ld(x,y)$	$\mathrm{Lip}(u) \le L$	Yes	No
$d_M^2/2$ on round spheres/products	Geometric c-concave potentials	Yes	No
Smooth general c	Convex iff NNCC, under FKM hypotheses	If NNCC	Usually no
Finite matrix C	Min-plus row span	Not generally	Tropical

Before asserting convexity or conicity, one should check:

1. Is the cost d or $d^2/2$? The first gives Lipschitz functions on every metric space; the second detects geometry.

2. Are the source and target domains full vector spaces, convex subsets, or compact manifolds? Restricted domains constrain supporting slopes.
3. Is “convex” ordinary or min-plus? Min-plus convexity is universal; ordinary convexity is not.
4. Does every potential have a nonempty c-subdifferential? On general spaces, the NNCC theorem concerns the supported subclass.
5. Does the cost have NNCC or only A3w? Only NNCC characterizes convexity of the whole potential class.
6. Is a cone claim compatible with the fixed Lipschitz/semiconcavity scale imposed by the cost? Usually it is not.

The central organizing principle is therefore:

Lower-envelope structure is universal; ordinary convexity is governed by NNCC; ordinary conicity requires an additional scaling symmetry.

References

Alessio Figalli, Young-Heon Kim, and Robert J. McCann. When is multidimensional screening a convex program? *Journal of Economic Theory*, 146(2):454–478, 2011. doi: 10.1016/j.jet.2010.11.006.

Alessio Figalli, Young-Heon Kim, and Robert J. McCann. Regularity of optimal transport maps on multiple products of spheres. *Journal of the European Mathematical Society*, 15(4): 1131–1166, 2013. doi: 10.4171/JEMS/388.

Anatole Gallouët, Quentin Mérigot, and Boris Thibert. Strong c-concavity and stability in optimal transport. *Journal de Mathématiques Pures et Appliquées*, 205:103773, 2026. doi: 10.1016/j.matpur.2025.103773.

Young-Heon Kim and Robert J. McCann. Continuity, curvature, and the general covariance of optimal transportation. *Journal of the European Mathematical Society*, 12(4):1009–1040, 2010. doi: 10.4171/JEMS/221.

Young-Heon Kim and Robert J. McCann. Towards the smoothness of optimal maps on riemannian submersions and riemannian products (of round spheres in particular). *Journal für die reine und angewandte Mathematik*, 664:1–27, 2012. doi: 10.1515/CRELLE.2011.105.

Flavien Léger, Gabriele Todeschi, and François-Xavier Vialard. Nonnegative cross-curvature in infinite dimensions: Synthetic definition and spaces of measures. *Geometric and Functional Analysis*, 35(6):1638–1711, 2025. doi: 10.1007/s00039-025-00725-x.

Grégoire Loeper and Neil S. Trudinger. Weak formulation of the MTW condition and convexity properties of potentials. *Methods and Applications of Analysis*, 28(1):53–60, 2021. doi: 10.4310/MAA.2021.v28.n1.a4.

R. Tyrrell Rockafellar. *Convex Analysis*. Princeton University Press, Princeton, NJ, 1970.

Tomonari Sei. A jacobian inequality for gradient maps on the sphere and its application to directional statistics. *Communications in Statistics—Theory and Methods*, 42(14):2525–2542, 2013. doi: 10.1080/03610926.2011.563017.

Cédric Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, 2009. doi: 10.1007/978-3-540-71050-9.